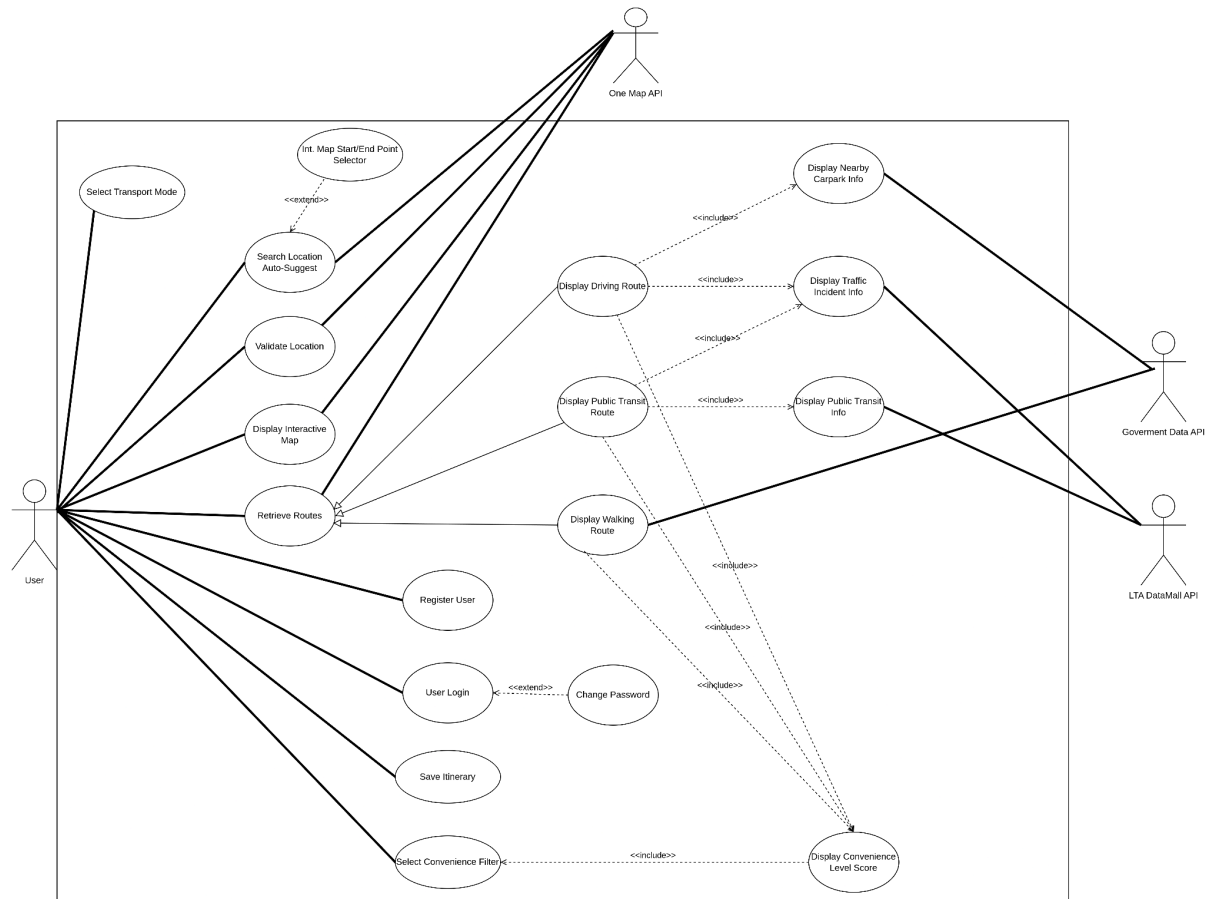


Use Case Model and Use Case Descriptions

Use Case Model



Use Case Descriptions

Use Case ID :	1.1		
Use Case Name:	Search Location Auto-Suggest		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User, OneMap API
Description	Enables users to search for Singapore locations using a search bar with auto-suggest functionality. The system will query the OneMap Search API based on the user's input and display a dropdown list of possible matching locations to facilitate selection of start and end points
Preconditions	1. The system has access to the OneMap Search API 2. The user is able to access the search bar interface
Postconditions	The system returns a list of auto-suggested locations based on the user's input via a dropdown
Priority	High
Frequency of Use	High
Flow of Events	1. The user enters a partial or full location name in the search bar (e.g. "Orchard" or "Changi Airport") 2. The system sends the user's input to the OneMap Search API in real-time as the user types 3. The OneMap Search API returns a list of matching Singapore locations based on the input 4. The system displays the list of locations in an auto-suggest dropdown below the search bar
Alternative Flows	AF-S3: No Matching Locations 1. If the OneMap Search API returns no results for the user's input, the system will display an empty auto-suggest dropdown
Exceptions	API Error: 1. If the OneMap Search API is unreachable, the system will display an error message: no suggestion will be displayed.
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.2		
Use Case Name:	Validate Location		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User, OneMap API
Description	Validates that the user's submitted start and end locations are valid Singapore locations by querying the OneMap Search API to retrieve their coordinates. The locations are provided as text input in a search bar, and validation ensures they correspond to recognisable Singapore addresses
Preconditions	1. The system has access to the OneMap Search API 2. The user has entered start and end locations as text in the search bar
Postconditions	1. The system confirms the start and end locations are valid and retrieves their coordinates 2. The system continues to use case 1.3
Priority	High
Frequency of Use	High
Flow of Events	1. The user submits the start and end locations entered as text in the search bar and clicks "fetch route" 2. The system sends the submitted location text to the OneMap Search API 3. OneMap Search API returns the corresponding coordinates for the respective start and end location if valid 4. The system stores the coordinates and accepts the locations as valid 5. The system proceeds to use case 1.3
Alternative Flows	AF-S3: Incomplete or Ambiguous Location 1. If the submitted location text is incomplete or ambiguous (e.g. "Orchard"), the OneMap Search API will return multiple locations and their coordinates instead of a single location. In this case, the system will choose the first location amongst all locations returned by the OneMap Search API
Exceptions	Invalid Location: 1. If the OneMap Search API returns no locations based on either the user's start and end location input, no location will be returned/auto completed.

	API Error: 1. If the OneMap Search API is unreachable, the system will display an error message: "Unable to validate locations. Please try again later."
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.3		
Use Case Name:	Select Mode of Transport		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	Allow the user to select between modes of transport
Preconditions	Start and end location validated
Postconditions	Display the information related to the mode of transport selected by the user
Priority	High
Frequency of Use	High
Flow of Events	1. User enters the main page which displays the multiple transport tabs (Best, Driving, Public Transport, Walking etc) 2. User clicks on the respective tab (mode of transport) 3. The system will query the external API (LTA data mall / onegov) for all modes of transport, regardless of the selected tab. 4. The system displays the results with respect to the selected mode of transport.

	5. Users can swap to other transport tabs and view 3 - 4 again.
Alternative Flows	AF-S3: Data Unavailable (from external API) 1. External API does not return any routes for a specific transport mode (network or invalid) 2. Tab will display error message "No routes available"
Exceptions	API Error: 1. The system cannot connect to the external API server. 2. The system will display an error message, "Connection Error."
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.4		
Use Case Name:	Select Convenience Filter		
Created by:	Benjamin Phang	Last updated by:	Benjamin Phang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	The system will display a drop down of filters which the user can select. These filters modify and directly affect convenience score in turn affecting routes displayed to the user.
Preconditions	Start and end location validated
Postconditions	1. The system saves the convenience score filter preference
Priority	High
Frequency of Use	High
Flow of Events	1. The system will display a drop down of filters selectable by the user (all filters are selected by default) 2. The user can select/deselect singular, multiple, or no convenience filters by adjusting a slider bar that saves filter preferences. 3. The system will preserve the filter preference across transport tabs.

	4. The system stores the user's convenience score filter preference for further calculations
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.5		
Use Case Name:	Display Convenience Level Score		
Created by:	Benjamin Phang	Last updated by:	Benjamin Phang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	The system will calculate and display a convenience level score for each mode of transport. The score will be within the range of 1-10, and they will be displayed to the user by rank based on the convenience level score
Preconditions	Start and end location validated
Postconditions	1. Convenience score displayed per route 2. "Best" tab indicating the top three routes by convenience score
Priority	High
Frequency of Use	High
Flow of Events	1. The system will retrieve the user's selected convenience filters 2. Convenience score is calculated for all routes within the range of 1 - 10 3. Convenience score is displayed to the user on each route card 4. The top three convenience scores from all transport modes are ranked in descending order in a separate "Best" tab
Alternative Flows	None

Exceptions	None
Includes	1.4
Special Requirements	None
Assumptions	External API provides accurate and updated route data
Notes and Issues	None

Use Case ID :	1.6		
Use Case Name:	Display Interactive Map		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, One Map API		
Description	User to be provided an interactive map interface which allows users to interact with such as being able to click on locations		
Preconditions	Users must have an internet connection to be able to connect to the application and API is working		
Postconditions	The interactive map interface has been generated and the user is able to see the map of Singapore		
Priority	High		
Frequency of Use	High		
Flow of Events	1. The user opens up the app. 2. Upon opening the application, the main UI will request for OneMapAPI data. 3. API will receive the request and return the main UI with map data. 4. The main UI will receive the data and display the full map of Singapore with the ability to interact with it.		
Alternative Flows	AF-S2: Interactive map is not displayed 1. The user opens the application but the interactive map does not load in, the user will reopen the application.		
Exceptions	API Error 1. The application goes to external API to retrieve info but is met with an error with the API 2. Displays an error message to inform the user that the API is currently experiencing troubles (API system not functional,		

	API unable to determine information given, etc.)
Includes	None
Special Requirements	None
Assumptions	Assumes that Application will only be used in Singapore
Notes and Issues	Locations are that not in the system of the external API will not have information on the interactive map

Use Case ID :	1.7		
Use Case Name:	Retrieve Routes		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, OneMap API
Description	This use case retrieves route options from the OneMap API based on the user's start and end location
Preconditions	1. The system has access to the OneMap API 2. Start and end location have been validated
Postconditions	The system has retrieved and prepared route options for the selected mode of transport.
Priority	High
Frequency of Use	High
Flow of Events	1. The system sends a request to the OneMap API with the coordinates of the start and end location 2. The OneMap API responds with route options 3. The system processes the response and prepares the route data for display. 4. The system continues to either use case 2.1, 3.1, 4.1
Alternative Flows	None
Exceptions	API Error: 1. If the OneMap API is unreachable, the system will display an error message: "Unable to retrieve routes. Please try again later."
Includes	None

Special Requirements	None
Assumptions	Assumes User has selected Starting Point and Destination
Notes and Issues	None

Use Case ID :	1.8		
Use Case Name:	Save Routes (Itinerary)		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User
Description	This interface will show the user their saved/favourited routes for ease of use if this is a common route they take
Preconditions	User is logged into their account and has chosen a Starting Point and Destination with the routes and itinerary generated
Postconditions	User has picked a route and saved it to Saved Routes interface
Priority	Low
Frequency of Use	Low
Flow of Events	1. After the user has displayed the generated itinerary, the user will click the 'Save' button located in the side bar. 2. Itinerary and filter will be saved to the Saved Routes interface for future reference or use 3. User will be able to access their saved route through the saved routes tab
Alternative Flows	AF-S3 No Route option saved 1. If the user did not save any routes previously, the system does not display any saved routes. 2. User to select a Starting Point and Destination 3. Users can click the 'Save' button to save the itinerary and filter values.

Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.9		
Use Case Name:	Interactive Map Start/End Point Selector		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	1/9/2025	Date Last Updated:	1/9/2025

Actor:	User
Description	User can pick their Starting Point or Destination by clicking directly on the interactive map
Preconditions	The map is displayed and clearly visible to the user
Postconditions	System receives start and end location data to be validated.
Priority	Low
Frequency of Use	Low
Flow of Events	1. User will click on a location on the interactive map 2. Information related to the location will pop up as a bubble (longitude and latitude) 3. Application will automatically fill in the respective position
Alternative Flows	AF-S1 Location is invalid point in the map 1. User clicks on a location which has no valid coordinate 2. Display Error Message
Exceptions	Interactive map API is not unreachable
Includes	1.1, 1.2, 1.6
Special Requirements	None
Assumptions	None

Notes and Issues	None
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Use Case ID:	2.1		
Use Case Name:	Display Driving Route		
Created By:	Dinh	Last Updated By:	25/08/2025
Date:	25/08/2025	Date Last Updated:	25/08/2025

Actor:	User, One Map API
Description	Display route results in driving mode.
Preconditions	User input start and end destination and select driving as preferred mode of transport.
Postconditions	The system shall display route results as a "route card"
Priority	Medium
Frequency of Use	High
Flow of Events	1. The user selects driving as the preferred mode of transport. 2. The system retrieves the list of available driving routes from the parent use case Retrieve Routes. 3. The system displays route cards for each driving route (e.g., including distance, travel time). 4. The system plots the driving routes on the interactive map. 5. The user may select a route card to view detailed instructions
Alternative Flows	AF-S3: 1. If there are no possible driving routes, the system displays "no routes available" in the driving tab.
Exceptions	None
Includes	1.5, 2.2, 2.3
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID:	2.2		
Use Case Name:	Display Nearby Carparks Info		
Created By:	Dinh	Last Updated By:	25/08/2025
Date:	25/08/2025	Date Last Updated:	25/08/2025

Actor:	User, Government Data API
Description	Display nearby carparks info in driving mode.
Preconditions	User input start and end destination and select driving as preferred mode of transport.
Postconditions	Information about carparks near to the end destination is displayed.
Priority	Medium
Frequency of Use	High
Flow of Events	1. The user selects driving as mode of transport. 2. The system gets driving routes by querying the external API. 3. The system retrieves possible routes from external API. 4. For each route, the systems retrieve information about nearby carparks such as availability. 5. The system displays nearby carparks as pins in the map . 6. If users click on the pins, a window pops up showing the detailed information about the carpark.
Alternative Flows	AF-S6: 1. If there is a lack of data about some data like parking rates, the system displays that the information is not available.
Exceptions	EX1: If the API is down, 1. The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	2.1
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	2.3		
Use Case Name:	Display Traffic Incident Info		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, LTA DataMall API
Description	Retrieve traffic incidents information on the route.
Preconditions	User input start and end destination.
Postconditions	Information about traffic incidents on the route is displayed.
Priority	Medium
Frequency of Use	High
Flow of Events	1. The system gets possible routes by querying the external API. 2. For each route, the system queries external API and filters incidents that directly impact the route. For driving mode, the system queries traffic accidents on the road. 3. The system displays that information on the map with pins on the affected route segments and names of roads affected on the route result card.
Alternative Flows	AF-S3: 1. No incident found, the system does not display any pins on the route.
Exceptions	EX1: If the API is down, 1. The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	2.1, 2.2
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	3.1
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Use Case Name:	Display Public Transit Route		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, One Map API
Description	Show buses/trains needed with transfer points.
Preconditions	User input start, end destination and select public transport mode.
Postconditions	Information about traffic incidents on the route is displayed.
Priority	High
Frequency of Use	High
Flow of Events	1. The user selects public transit as the preferred mode of transport. 2. The system retrieves the list of available public transit routes from the parent use case Retrieve Routes. 3. The system displays route cards for each public transit route (e.g., including distance, travel time). 4. The system plots the public transit routes on the interactive map. 5. The user may select a route card to view detailed instructions such as buses and MRT to take as well as walking segments
Alternative Flows	AF-S2: 1. Services are unavailable due to the time of day, then the system displays that error to the user. AF-S1: 1. User input location outside of Singapore, the system then displays the message to the user.
Exceptions	None
Includes	1.5, 2.3, 3.2
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	3.2		
Use Case Name:	Display Public Transit Info		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, LTA DataMall API
Description	Display detailed information about platforms and crowd level.
Preconditions	User input start, end destination and select public transport mode.
Postconditions	Information is displayed to the user.
Priority	High
Frequency of Use	High
Flow of Events	1. User select a particular itinerary in the public transit mode. 2. For each route in the itinerary: a. System retrieves scheduled departure/arrival. b. System retrieves real-time updates such as platform density, average bus waiting time (where applicable) and estimated travel time. 3. API returned itineraries containing buses, MRT and walking segments.
Alternative Flows	AF-S2: 1. Platform information is missing, the system shows the information is not available.
Exceptions	EX1: If the API is down, 1. The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	4.1		
Use Case Name:	Display Walking Route		
Created by:	Andy	Last Updated By:	Andy
Date	26/8/2025	Date Last Updated:	26/8/2025

Actor:	User, OneMap API, Government Data API
Description	Shows Walking route
Preconditions	User input start and end destination and select walking as preferred mode of transport.
Postconditions	The system shall display walking route results as a “route card”
Priority	High
Frequency of Use	High
Flow of Events	1. The user selects public walking as the preferred mode of transport. 2. The system retrieves the list of available walking routes from the parent use case Retrieve Routes, and the forecast. 3. The system displays route cards for each walking route (e.g., including distance, travel time, weather forecast...). 4. The system plots the walking routes on the interactive map. 5. The user may select a route card to view detailed instructions on the walking directions
Alternative Flows	None
Exceptions	None
Includes	1.5
Special Requirements	None
Assumptions	Start and End location is validated
Notes and Issues	None

Use Case ID :	5.1		
Use Case Name:	User Login		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	The back and front end system involved in managing authentication services within the application. It is used in validating entered credentials and providing appropriate error messages otherwise
Preconditions	1. User must already have an existing account within application database 2. The system is already connected to the account database
Postconditions	1. User is assigned a login session token by the authentication service 2. The user is displayed an error message upon input of invalid credentials not present in the application's database
Priority	High
Frequency of Use	High
Flow of Events	1. The user opens the website. 2. The system pops up a login form with email and password and a login 3. The system validates the input credentials (e.g valid email, password length) 4. The system queries and validates with the account database to check account validity 5. If the account is valid, the system provides a login session token and redirects the user to the main page.
Alternative Flows	None

Exceptions	1. Invalid Login Credentials - If the user provides invalid login credentials, an error message with the text “invalid login credentials, please try again” message below the associated fields. 2. Attempt Lockout - After 5 failed login attempts, the account is temporarily locked and an email is sent to the user's email on suspicious login activity.
Includes	5.2
Special Requirements	1. The application in production must support HTTPS and have valid authority certificates. 2. The password in transit and in storage must be salted and hashed for data security
Assumptions	1. The user has an existing valid email
Notes and Issues	Optionally implement Google / external OAuth for login instead of creating an account with our system

Use Case ID :	5.2		
Use Case Name:	Register User		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	(Accountless) User
Description	Registration of a user with no existing account, providing credentials such as email address, password and username for added features of the app.
Preconditions	1. User is not logged in 2. User has an existing reachable email address
Postconditions	1. Account creation successful, updated in the system's credential database 2. Data stored is validated and sanitized to prevent against cross site scripting (XSS) 3. Confirmation email sent to email address
Priority	High
Frequency of Use	High
Flow of Events	1. User clicks on sign up button on the login screen 2. System will pop up a registration form with the following

	details - First Name - Last Name - Email Address - Password 3. System validates entered data with the following criteria a. Password i. 8 Characters ii. At least 1 numerical digit iii. At least 1 special symbol / character b. Email Address i. Presence of @ symbol 4. System sanitises and prevents encoding of HTML tags 5. Upon successful validation, create a new user and store it within the system's account database 6. The system displays "Account Creation Successful" and redirects the user to the main page.
Alternative Flows	AF-S3 Weak / Invalid Password / email - User inputs invalid password / email - System displays an error message below the respective field AF-S3 Account Exists - User inputs an email that already exists in the database. - Upon pressing login, the system displays an error.
Exceptions	Invalid Characters - Reject html encoded tags such as <script> and provide an error message for illegal use of characters
Includes	None
Special Requirements	1. The application in production must support HTTPS and have valid authority certificates. 2. The password in transit and in storage must be salted and hashed for data security 3. The system must sanitise and encode all user inputs to prevent XSS. 4. Error messages must not reveal anything about the system
Assumptions	1. The user has an existing valid email
Notes and Issues	Consider using third party oauth like Google instead of directly registering an account.

Use Case ID :	5.3		
Use Case Name:	Change Password		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	Allow users to change their password under their profile page
Preconditions	1. User must have an existing account and logged in to the system 2. The system is already connected to the account database
Postconditions	3. The user's password is updated in the database
Priority	High
Frequency of Use	High
Flow of Events	1. User presses the 'Change Password' button under their profile page. 2. User keys in their current password as well as their new password. 3. New password must meet password requirements and must match, otherwise an error message is returned. 4. The system updates the user's account in the database with the updated password.
Alternative Flows	None
Exceptions	1. Authentication Services Down - Provide an error code and display the message "Authentication Services are unreachable, please try again later" message 2. The domain of the sending email (email service that sends the reset link) must be of a unique and trusted domain. 3. Invalid Characters / Not meeting minimum requirements - Reject HTML-encoded tags such as <script> and provide an error message for illegal use of characters.
Includes	5.1
Special	None

Requirements	
Assumptions	1. The user has an existing valid account in the database
Notes and Issues	None